
Upward Bound Math and Science Center Drone Design

Summer 2025

*As the instructor for this course, I reserve the right to adjust this schedule
in any way that serves the educational needs of the students enrolled in this course.*

– Apar Pokhrel

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Instructor Information

Instructor Name

Apar Pokhrel

Office Hours

By appointment only (Teams/Canvas/Study Hall)

Email Address

utabms@gmail.com

Communication Guidelines

My preferred communication method is email.

Course Information

Time and Place of Class Meetings

Tuesday/Friday 2:40 – 4:10 PM Business Building (COBA) 140

[View Campus Map](#)

Time Zone

This course operates on Central Time. All times listed for class meeting times, exams, and assignment deadlines are in Central Time.

Description of Course Content

This course is geared towards students' interest in drone technology, combining principles of engineering, physics, and computer science. Students will learn how drones are designed, built, and programmed. The course will also focus on real-world applications in fields such as environmental science, public safety, delivery systems, and aerial photography.

There will also be an opportunity to use drones in a safe location towards the end of the class.

Student Learning Outcomes

By the end of this course, students will be able to:

1. **Explain the basic principles of flight** and how drones achieve lift, stability, and control through aerodynamics and propulsion systems.
2. **Identify and describe key components of a drone**, including motors, ESCs (electronic speed controllers), propellers, batteries, and flight controllers.
3. **Safely assemble and wire** a basic drone, demonstrating proper use of tools and understanding of electronic circuits.
4. **Understand FAA safety guidelines** and drone regulations, and apply responsible practices for drone operation.
5. **Work collaboratively** in design teams, demonstrating communication and effective problem-solving.
6. **Explore applications of drone technology** in the areas of environmental science, agriculture, disaster responses, and package delivery among others.

Textbooks and Other Course Materials

Required Textbooks and Materials

No required textbook is needed for this course. All reading materials, circuit designs and code examples will be provided by the instructor or made accessible through Canvas/GitHub.

Code examples and in-class demonstrations will be uploaded to GitHub. Check out [UBMS-Drone-Design](#) for updates.

Expectations for Out-of-Class Study

Students should expect to spend additional time outside of class on activities such as reading materials, completing assignments, and preparing for exams.

In-Class Expectations

Students should adhere to the guidelines in the Student Handbook, respect classmates and the instructor, practice proper internet etiquette. Computers and cell phones should be muted when not in use and not visible or seen.

Grading Information

Graded Assignments & Values

Assignment Name	Value (%)
Participation	10
Team Design & Presentation	25
Drone Application Case Study	10
Drone Simulation	15
Mid-term Quiz	20
Post-test	20
Total	100

Students are expected to keep track of their performance throughout the semester which Canvas facilitates and seek guidance from available sources (including the instructor) if their performance drops below

Range (pts or %)	Letter Grade
[86.5 – 100]	A
[74.5 – 86.5)	B
[64.5, 74.5)	C
[54.5, 64.5)	D
[0, 54.5)	F

Make-Up Exams & Late Work Policy

Make-up exams will be permitted if a class period for the exam is cancelled or under unusual circumstances with prior notification from the student.

Grade Grievances

Any appeal of a grade must be made by contacting the instructor and following the procedures and deadlines set by the Upward Bound Math & Science program.

University & Course Policies

UTA students are encouraged to review the institutional policies and informational sections below and reach out to the specific office with any questions. To view this institutional information, please visit the [Institutional Information](https://resources.uta.edu/provost/course-related-info/institutional-policies.php) page (<https://resources.uta.edu/provost/course-related-info/institutional-policies.php>), which includes the following policies, among others:

- Drop Policy: Refer to the Director of the Upward Bound Math & Science Center and the Student Handbook.
- Disability Accommodations
- Title IX Policy
- Academic Integrity

Attendance

Attendance is mandatory. Notification of absences must be sent via e-mail to the Upward Bound

Symposium and Awards Banquet

Math & Science Symposium will be held on July 7, 3:00 -5:00 pm. This gives you an opportunity to present your research and showcase your accomplishments! Family members, instructors, and university staff are invited to attend.

Awards Banquet and Ceremony will be held on July 7, 6:00 – 9:15 pm to recognize the achievements of the summer participants.

Generative AI Use in This Course

The use of Generative AI (GenAI) in course assignments and assessments must align with the guidelines established by the instructor. Unauthorized use of GenAI could result in breaches of academic integrity. Instructors bear the responsibility of clearly delineating the permissible uses of GenAI in their courses, underscoring the importance of responsible and ethical application of these tools.

The [UTA Office of Community Standards](#) articulates the university's stance on [academic integrity and scholastic dishonesty](#). These standards extend to the use of GenAI. Unauthorized or unapproved use of GenAI in academic work falls within the scope of these policies and will be subject to the same disciplinary procedures.

As the instructor of this course, I have adopted the following policy on Student use of GenAI:

Restricted Use of GenAI

Approach	Description
Restricted Use of GenAI	While this course recognizes the potential benefits of Generative AI (GenAI) as a supplementary tool for certain learning activities, its use is restricted to specific assignments where GenAI's role is clearly defined and aligns with the course's Student Learning Outcomes (SLOs). These assignments will be clearly marked, and students must adhere to the guidelines provided for GenAI use. Unauthorized use of GenAI outside these parameters will be considered a breach of academic integrity.

Academic & Wellness Resources

Academic Success

Students can access STEM Halls for help. Students can also utilize individual study time Monday-Thursday 1:30- 2:30 pm to work on projects. Instructor will be available through individual appointments if needed.

The English Writing Center (411LIBR)

The Writing Center offers **FREE** tutoring in 15-, 30-, 45-, and 60-minute face-to-face and online sessions to all UTA students on any phase of their UTA coursework. Register and make appointments online at the [Writing Center](https://uta.mywconline.com) (<https://uta.mywconline.com>). Classroom visits, workshops, and specialized services for graduate students and faculty are also available. Please see [Writing Center: OWL](http://www.uta.edu/owl) (<http://www.uta.edu/owl>) for detailed information on all our programs and services.

Academic Plaza

The Library's 2nd floor [Academic Plaza](http://library.uta.edu/academic-plaza) (<http://library.uta.edu/academic-plaza>) offers students a central hub of support services, including IDEAS Center, University Advising Services, Transfer UTA and various college/school advising hours. Services are available during the [library's hours](https://library.uta.edu/hours) (<https://library.uta.edu/hours>) of operation.

UTA CARE Team

NOTE: If a person's behavior poses an immediate threat to you or someone else, contact UTA Police at 817-272-3303 or dial 911. If you or someone you know needs to speak with a crisis counselor, please reach out to the [MAVS TALK 24-hour Crisis Line](https://www.uta.edu/student-affairs/caps/crisis) (<https://www.uta.edu/student-affairs/caps/crisis>) at 817-272-8255 or the [National Suicide and Crisis Lifeline](https://988lifeline.org/) (<https://988lifeline.org/>) at 988.

Librarian to Contact

Each academic unit has access to [Librarians by Academic Subject](https://libraries.uta.edu/research/librarians) (<https://libraries.uta.edu/research/librarians>) that can assist students with research projects, tutorials on plagiarism, and citation references, as well as support with databases and course reserves.

Safety Information & Resources

Face Covering Policy

Face coverings are not mandatory; all students and instructional staff are welcome to wear face coverings while they are on campus or in the classroom.

Emergency Exit Procedures

Should we experience an emergency event that requires evacuation of the building, students should exit the room and move toward the nearest exit, which is located to your right. ([Evac B 140.pdf](#)). When exiting the building during an emergency, do not take an elevator but use the stairwells instead. Faculty members and instructional staff will assist students in selecting the safest route for evacuation and will make arrangements to assist individuals with disabilities.

MavAlert System

The MavAlert system sends information to cell phones or email accounts of subscribed users in case of an emergency. Anyone can subscribe to MavAlerts at [Emergency Communication System](https://www.uta.edu/uta/emergency.php) (<https://www.uta.edu/uta/emergency.php>).

Emergency Phone Numbers

In case of an on-campus emergency, call the UT Arlington Police Department at **817-272-3003** (non-campus phone), **2-3003** (campus phone). You may also dial 911. Non-emergency number 817-272-3381

Course Schedule

Class Date(s)	Topic(s)	Activities	Assignments
June 3	Course Introduction & Pre-test	- Course overview & syllabus - Course Expectations - Pre-test	Pre-test assigned Team Design & Presentation assigned
June 6	Real-World Drone Applications	-Interactive session on drone applications -FAA guidelines & safety discussion - Introduction to Drone Safety	
June 10	Drone Components & Functions	- Intro to drone components (motors, propellers, frame, flight controller) - Intro to basic drone building concepts and flight principles	Drone Application Case Study Due
June 13	Drone Assembly & Programming	-prototyping & builds	
June 17			
June 20			Mid-term Quiz assigned
June 24			Drone Simulation Due
June 27	Final Build & Reflection	-Review builds & parts - Q&A - Final Reflections - Personal project ideas	
July 1	Outdoor Flights & Calibration	- Calibrate drone and test outdoor flight with builds - Record videos/logs for review - Guest speaker drone flight test	
July 4	No classes		Post-test assigned (July 3) Team Design & Presentation due

Class Date(s)	Topic(s)	Activities	Assignments
July 7	Math & Science Symposium Awards Banquet & Ceremony		